“Well,” said Alice, “mostly we have to say, too bad for them. Stuff happens. But the total weight assigned to such observers is so small that we shouldn’t worry about them too much. Not to mention that, after they get fifty spin-ups in a row, the next fifty trials will still map onto the Born-rule predictions with overwhelming probability. Most likely they will attribute their original lucky streak to experimental error, and have a fun story to tell their lab mates. It’s just like a classical universe that is just really big. If conditions that we see in the universe around us continue infinitely far in every direction, it is overwhelmingly probable that there are other civilizations just like ours—an infinite number, in fact—doing experiments to test quantum mechanics. Even if each of them is likely to see Born-rule probabilities, given that there are an infinite number of them, some of them will see very different statistics. In that case they may be led to draw incorrect conclusions about how quantum mechanics works. Those observers would be unlucky, but we can take consolation in the fact that they are also very infrequent among the set of all observers in the universe.”

“Small consolation for them! In your view of physics, there will always be observers out there who get the laws of nature utterly wrong.”

“Nobody ever promised them a rose garden. That worry exists in any theory where there are sufficiently large numbers of observers; Many-Worlds is just one example of such a theory. The point is that in Everettian quantum mechanics, there is a way to compare all of the different worlds: take the amplitudes of their branches, and square them. The branches in which very surprising things happen have very, very tiny amplitudes. They are rare in the set of all worlds. We shouldn’t be any more bothered by their existence than we are by unlucky observers in infinitely large universes.”

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“Not sure I’m convinced here, but let’s just enter my worry into the record and move on.” He squinted at the list of questions he had brought up on his phone. “I’ve been doing a bit of reading—even some of your papers—and one thing I do appreciate about Many-Worlds is that it removes any lingering mystery about when a measurement takes place. There’s nothing special about measurement; it’s just when a quantum system that’s in a superposition becomes entangled with the larger environment, leading to